USS, OMVS, iShell, and Batch

Writing shell scripts and running under USS and batch


TEKsystems
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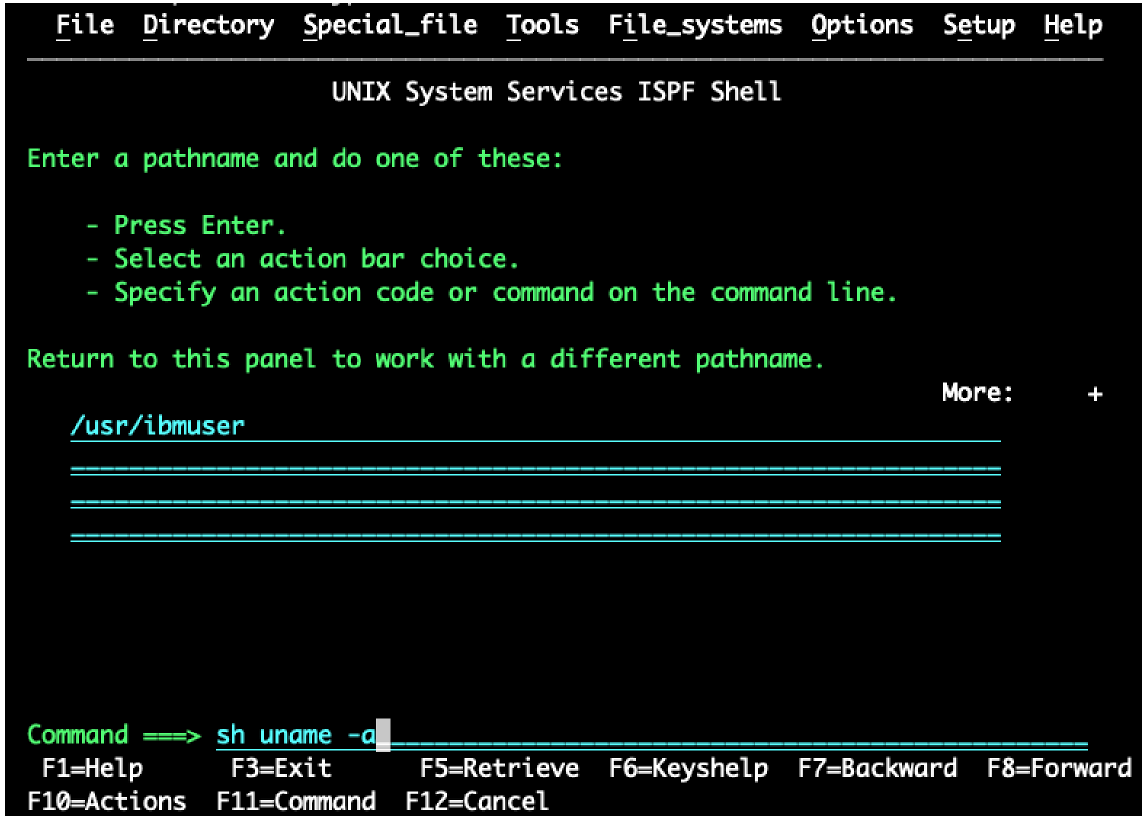
USS

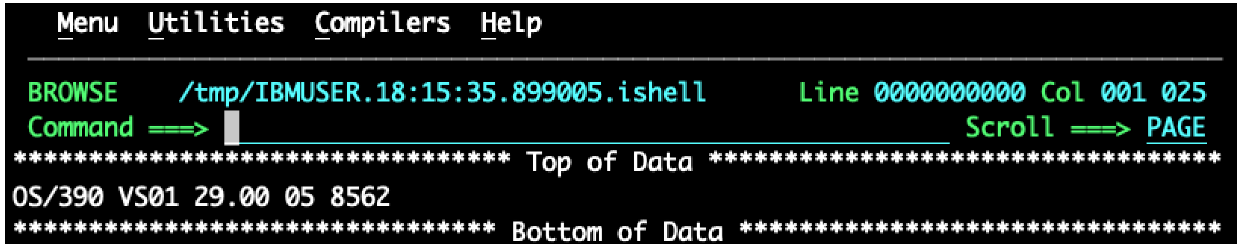
USS, or Unix System Services, is the brand name for POSIX functionality in the z/OS operating system.

You can access USS in any of three ways:

- From TSO or ISPF Option 6 using iShell (shown at right)
- From TSO of ISPF Option 6 using omvs
- From a secure shell (ssh) connection



iShell is an ISPF-style wrapper around USS commands. It's meant to be a friendly starting point for people with experience working in TSO/ISPF who are new to a Unix-style command line interface.



Those who are already familiar with Unix and Linux usually find iShell a bit clunky and counterintuitive.

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- From TSO of ISPF Option 6 using omvs (shown at right)
- From a secure shell (ssh) connection shown below right)

```
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IBMUSER : /u/ibmuser : > uname -a
OS/390 VS01 29.00 05 8562
IBMUSER : /u/ibmuser : >
```

These methods give you a more normal-looking command-line interface. By default, USS doesn't ship with all the nice shells and tools most Unix and Linux systems have out of the box, but you can install them.

uname reports the OS as "OS/390". It's the same as z/OS.

```
IBMUSER : /u/ibmuser : > uname -a
OS/390 VS01 29.00 05 8562
IBMUSER : /u/ibmuser : >
```

USS

```
IBMUSER : /u/ibmuser : > uname -a
OS/390 VS01 29.00 05 8562
IBMUSER : /u/ibmuser : > cp "//'ibmuser.lab.rexx(howdy)'" howdy.sh
IBMUSER : /u/ibmuser : > ls howdy.sh
howdy.sh
IBMUSER : /u/ibmuser : > cat howdy.sh
/* REXX */                                                        00010000
say "Hello, World!"                                               00020000
exit 0                                                            00030000
IBMUSER : /u/ibmuser : >
```

Since USS is built into z/OS and isn't really a separate Unix instance, it has access to MVS data sets as well as USS filesystems.

The example shows a cp (copy) command that copies a PDS member to a USS file. You can see that basic Unix/Linux commands are supported.

To refer to MVS assets here, you have to wrap their names in special characters. The entire command line is enclosed in quotation marks, and the data set name must be prefixed with two slashes. After the two slashes, the MVS data set name has to be enclosed in apostrophes. It doesn't have to be capitalized.

Notice that the cp command picked up the sequence numbers in columns 73-80 of the PDS member. USS doesn't know what that means, and just treats all 80 bytes as data. Use ISPF unnum and number off to get rid of the sequence numbers, if you will need to copy the data set or member to USS.

USS

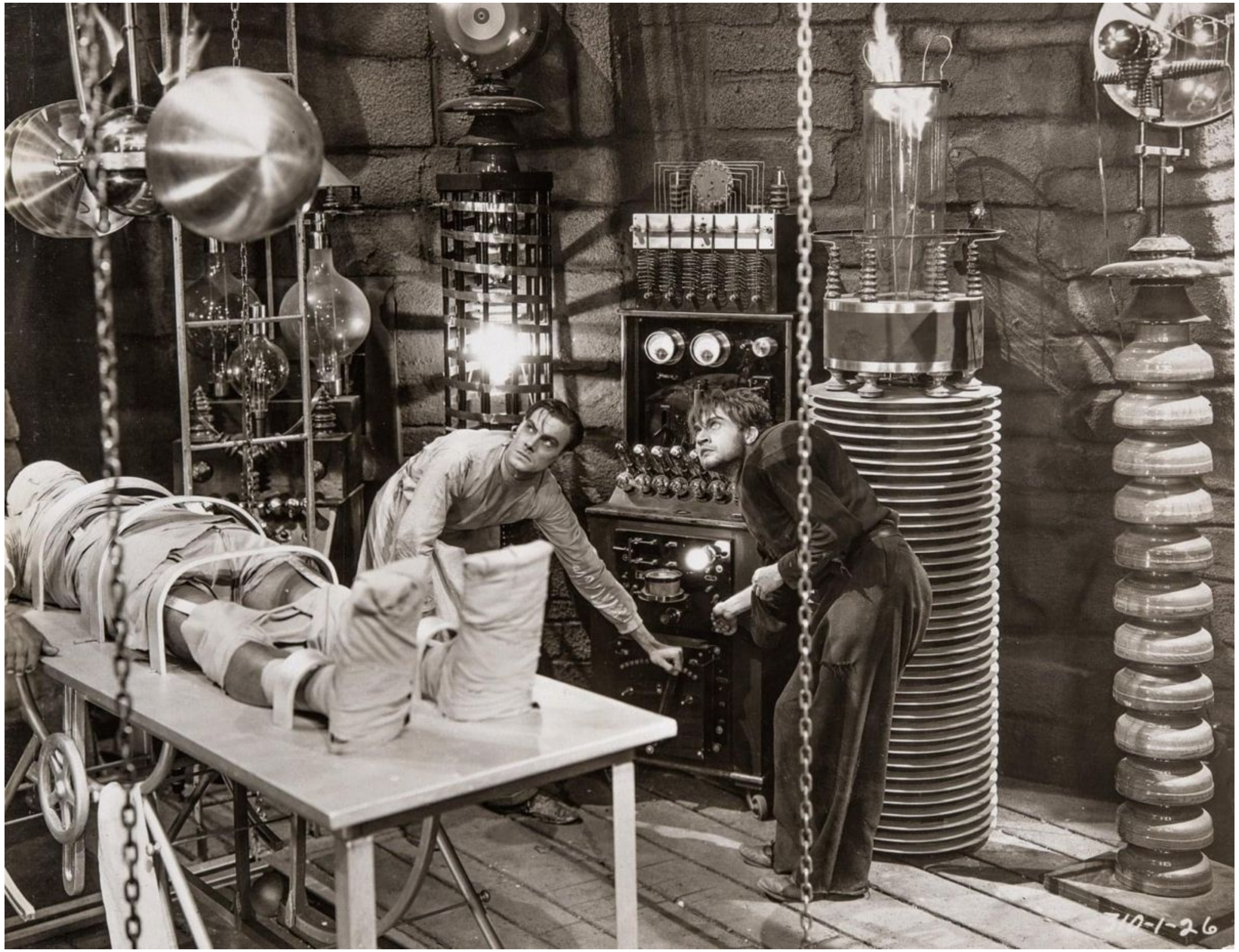
Let's edit the file we copied and make it a shell script instead of a REXX script.

```
IBMUSER : /u/ibmuser : > cat howdy.sh
/* REXX */
say "Hello, World!"
exit 0
IBMUSER : /u/ibmuser : > vi howdy.sh
```

```
#!/bin/sh
echo "Hello, World!"
exit 0
~
```

```
~
:wq
IBMUSER : /u/ibmuser : > chmod +x howdy.sh
IBMUSER : /u/ibmuser : > ./howdy.sh
Hello, World!
IBMUSER : /u/ibmuser : >
```

LAB Z/OS 22: WRITING A USS SHELL SCRIPT



USS BATCH

Now that you have a shell script and you know it works, it's time to run it in batch mode using program BPXBATCH.

WHAT YOU KNOW AND WHAT YOU CAN DO

- You can write a simple USS shell script.
- You can run a shell script from a USS command line.
- You can run a shell script using the BPXBATCH program.

WHAT'S NEXT

- z/OS Address Spaces, Addressing Modes, LPARs, SYSPLEX